MIL-S-19500/13B
26 January 1965
SUPERSEDING
MIL-T-19500/13A
8 January 1958

MILITARY SPECIFICATION

# SEMICONDUCTOR DEVICE, TRANSISTOR, PNP, GERMANIUM, HIGH-POWER

## TYPE 2N174A

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers the detail requirements for a high-power, PNP germanium transistor and is in accordance with MIL-S-19500 except as otherwise specified herein.

1.2 Physical dimensions. See figure 1 (TO-6).

1.3 Absolute-maximum ratings:

| $P_C$ 1/ $T_{MB}$ = 25°C | $I_E$ | $I_B$ | $V_{CBO}$ | $V_{EBO}$ | $V_{CEO}$ | $T_{stg}$ |
|---|---|---|---|---|---|---|
| (W) | (Adc) | (Adc) | (Vdc) | (Vdc) | (Vdc) | (°C) |
| 75 | 14 | 4 | -80 | -60 | -40 | -65 to +100 |

1/ Derate 1.0 W/°C for $T_{MB} \geq 25$°C.

1.4 Primary electrical characteristics.

| | $h_{FE}$ at $V_{CE}$ = -2 Vdc | | $V_{CE}(sat)$ $I_C$ = -12 Adc $I_B$ = -2 Adc | $f_{hfb}$ | $\theta_{J-C}$ |
|---|---|---|---|---|---|
| | $I_C$ = -1.2 Adc | $I_C$ = -5 Adc | | | |
| | | | (Vdc) | (mc) | (°C/W) |
| Min | 40 | 25 | --- | 0.1 | --- |
| Max | 80 | --- | 0.7 | --- | 0.5 |

## 2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATION

Military

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

Military

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 5960

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.

3. REQUIREMENTS

3.1 General. Requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. The abbreviations, symbols, and definitions used herein are defined in MIL-S-19500 and as follows:

$T_{MB}$ . . . . . Mounting-base temperature.

3.3 Design and construction. Transistors shall be of the design, construction, and physical dimensions specified on figure 1.

3.4 Performance characteristics. Performance characteristics shall be as specified in tables I, II, and III and as follows:

3.4.1 Barometric pressure, reduced (altitude operation). Transistors shall operate satisfactorily without voltage derating (see table III).

3.5 Marking. The following marking, specified in MIL-S-19500, may be omitted from the body of the transistor:

(a) Country of origin.
(b) Manufacturer's identification.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.2 Qualification inspection. Qualification inspection shall consist of the examinations and tests specified in tables I, II, and III.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of groups A, B, and C inspections.

4.3.1 Group A inspection. Group A inspection shall consist of the examinations and tests specified in table I.

4.3.2 Group B inspection. Group B inspection shall consist of the examinations and tests specified in table II.

4.3.3 Group C inspection. Group C inspection shall consist of the examinations and tests specified in table III. This inspection shall be conducted on the initial lot and thereafter every six months during production.

4.4 Methods of examination and test. Methods of examination and test shall be as specified in tables I, II, and III and as follows:

4.4.1 Salt spray (corrosion). The transistor shall be examined for evidence of corrosion and legibility of marking before the specified measurements are made (see table II).

4.4.2 Inspection conditions. All measurements are to be made at $T_C$ = 25°C unless otherwise specified.

4.4.3 Test measurement. Test measurement shall be made after thermal equilibrium has been reached at the temperature specified.

4.4.4 Pulse testing. Pulse measurements shall be used in accordance with Section 4 of MIL-STD-750.

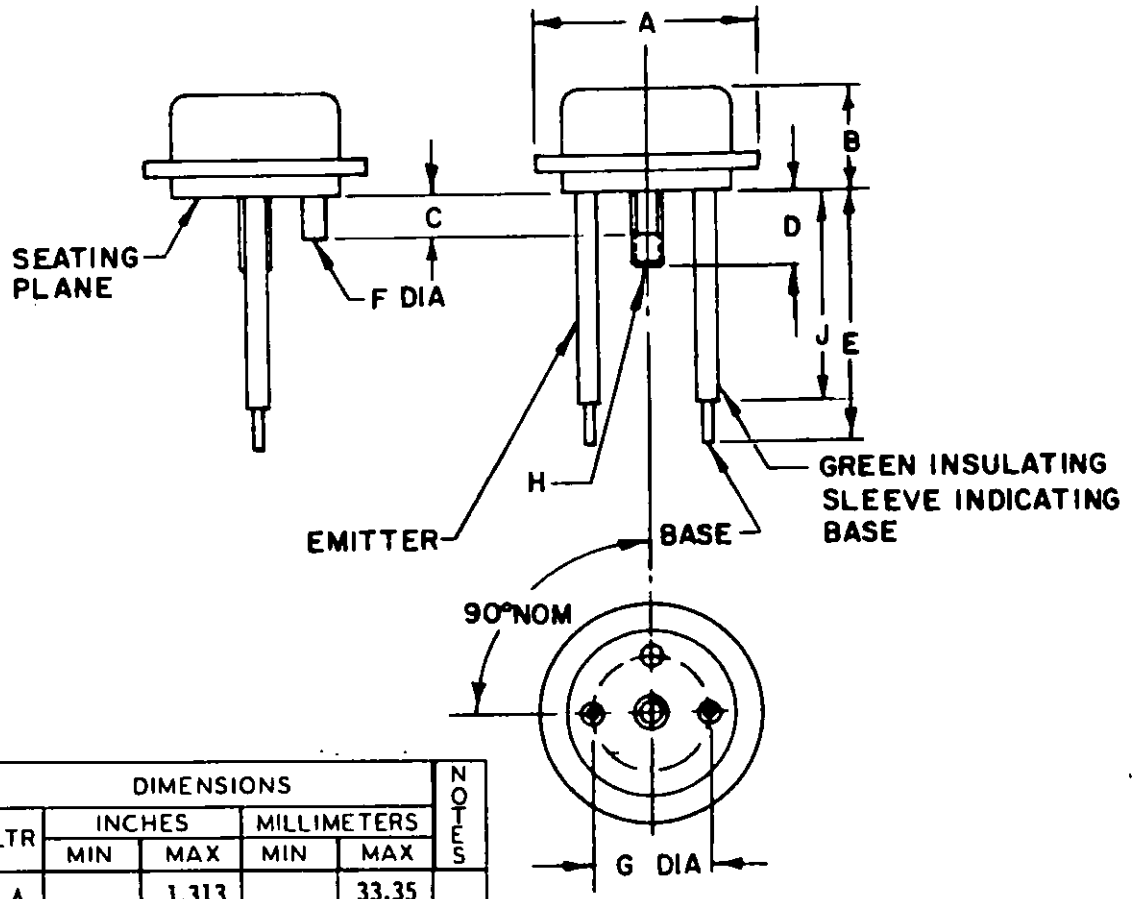
4.4.5 Solderability. The solderability test shall apply to the two flexible leads only. The insulating sleeve shall be removed during this test. The depth of immersion is to be within 1 inch of the seating plane. Acceptance criteria shall be that the determination is 95% covered by a continuous new solder coating to within 1-1/4 inch ± 1/8 inch of the seating plane

4.4.6 Moisture resistance. The insulation may be removed from the emitter and base leads prior to making the end-point measurement.

4.4.7 Terminal strength (stud torque). Acceptance criteria after the stud torque test shall be dimensions D and H of figure 1. Elongation will be allowed within these limits.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.




| DIMENSIONS | | | | | NOTES |
|---|---|---|---|---|---|
| LTR | INCHES | | MILLIMETERS | | |
| | MIN | MAX | MIN | MAX | |
| A | | 1.313 | | 33.35 | |
| B | | .625 | | 15.88 | |
| C | .100 | .312 | 2.54 | 7.92 | |
| D | .375 | .500 | 9.53 | 12.70 | |
| E | 1.500 | | 38.10 | | 4 |
| F | | .141 | | 3.58 | |
| G | .670 | .710 | 17.02 | 18.03 | |
| H | | | | | 3,5 |
| J | 1.125 | 1.375 | 28.58 | 34.93 | 4,6 |

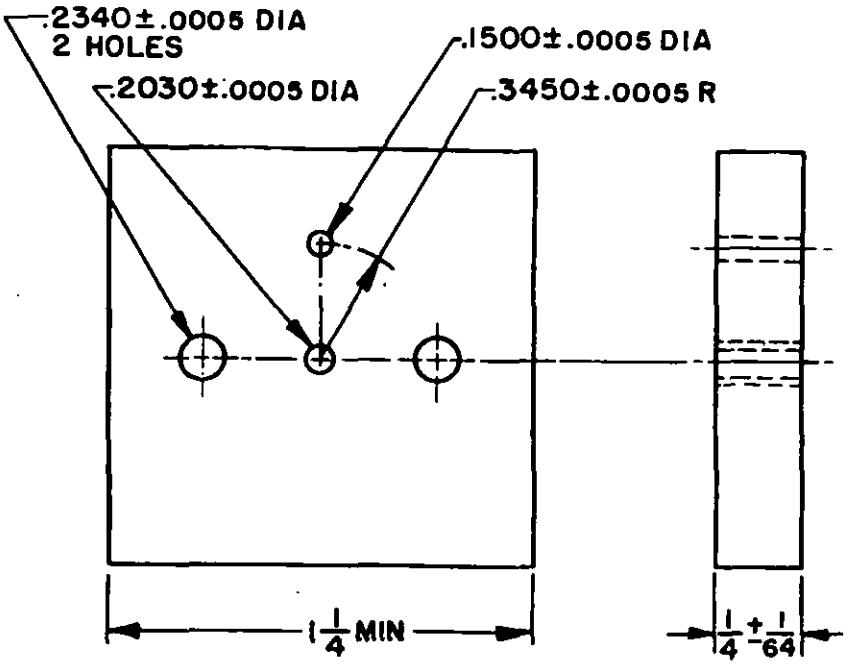
NOTES:

1. All dimensions in inches.
2. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.
3. 10-32 UNF-2A.
4. Two leads.
5. The collector shall be internally connected to the mounting base.
6. Measured from the seating plane.
7. A preferred alinement gage is shown on figure 2.

Figure 1. Physical dimensions of transistor type 2N174A.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.




| INCHES | MM |
|---|---|
| .0005 | .01 |
| 1/64 | .40 |
| .1500 | 3.81 |
| .2030 | 5.16 |
| .2340 | 5.94 |
| 1/4 | 6.35 |
| .3450 | 8.76 |
| 1-1/4 | 31.75 |

NOTES:
1. Metric equivalents (to the nearest .01 mm) are given for general information only and are based upon 1 inch = 25.4 mm.

Figure 2. Alinement gage for transistor type 2N174A.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.

Table I. Group A inspection.

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 1 | | | 10 | | | | |
| Visual and mechanical examination | 2071 | | | | | | |
| Subgroup 2 | | | 5 | | | | |
| Breakdown voltage, emitter to base | 3026 | Bias cond. D; $I_E$ = -10 mAdc | | $BV_{EBO}$ | -60 | --- | Vdc |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -2 Vdc | | $I_{CBO}$ | --- | -200 | uAdc |
| Breakdown voltage , collector to base | 3001 | Bias cond. D; $I_C$ = -15 mAdc | | $BV_{CBO}$ | -80 | --- | Vdc |
| Floating potential | 3020 | $V_{CB}$ = -80 Vdc; Voltmeter input resistance = 10 Meg. min. | | $V_{EBF}$ | --- | -1.0 | Vdc |
| Collector to emitter voltage (saturated) | 3071 | $I_C$ = -12 Adc; $I_B$ = -2 Adc | | $V_{CE(sat)}$ | --- | -0.7 | Vdc |
| Base emitter voltage (nonsaturated) | 3066 | Test cond. B; $V_{CE}$ = -2 Vdc; $I_C$ = -5 Adc | | $V_{BE}$ | --- | -0.9 | Vdc |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -2 Vdc; $I_C$ = -1.2 Adc; pulsed (see 4.4.4) | | $h_{FE}$ | 40 | 80 | --- |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -2 Vdc; $I_C$ = -5 Adc; pulsed (see 4.4.4) | | $h_{FE}$ | 25 | --- | --- |
| Breakdown voltage, collector to emitter | 3011 | Bias cond. D; $I_C$ = -250 mAdc | | $BV_{CEO}$ | -40 | --- | Vdc |
| Subgroup 3 | | | 10 | | | | |
| Small-signal short-circuit forward-current transfer-ratio cutoff frequency | 3301 | $V_{CE}$ = -12 Vdc; $I_C$ = -1.2 Adc | | $f_{hfb}$ | 0.1 | --- | mc |
| High-temperature operation: | | $T_{MB}$ = +71°C min | | | | | |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -30 Vdc | | $I_{CBO}$ | --- | -6 | mAdc |



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection.

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 1 | | | 20 | | | | |
| Physical dimensions | 2066 | (See figure 1) | | | | | |
| Subgroup 2 | | | 15 | | | | |
| Solderability | 2026 | (See 4.4.5) | | --- | --- | --- | --- |
| Temperature cycling | 1051 | Test cond. B, except step 3, $t_{max}$ = +95 $^{+5}_{-0}$ °C | | --- | --- | --- | --- |
| Thermal shock (glass strain) | 1056 | Test cond. B | | --- | --- | --- | --- |
| Moisture resistance | 1021 | Omit initial conditioning (see 4.4.6) | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -80 Vdc | | $I_{CBO}$ | --- | -15 | mAdc |
| Emitter to base cutoff current | 3061 | Bias cond. D; $V_{EB}$ = -60 Vdc | | $I_{EBO}$ | --- | -10 | mAdc |
| Subgroup 3 | | | 15 | | | | |
| Shock | 2016 | Nonoperating; 1500 G, approx. 0.5 msec; 5 blows in each orientation: $X_1$, $Y_1$, $Y_2$, and $Z_1$ | | --- | --- | --- | --- |
| Vibration fatigue | 2046 | Nonoperating | | --- | --- | --- | --- |
| Vibration, variable frequency | 2056 | Nonoperating | | --- | --- | --- | --- |
| Constant acceleration | 2006 | 2000 G, in each orientation: $X_1$, $Y_1$, $Y_2$ and $Z_1$ | | --- | --- | --- | --- |
| End points: (Same as subgroup 2) | | | | | | | |
| Subgroup 4 1/ | | | 15 | | | | |
| Terminal strength (tension) | 2036 | Test cond. A; 2 lbs.-10 sec. | | --- | --- | --- | --- |
| Terminal strength (stud torque) | 2036 | Test cond. D2; 12 in.-lbs. t = 30 sec. (see 4.4.7) | | --- | --- | --- | --- |
| Subgroup 5 | | | 20 | | | | |
| Salt spray (corrosion) | 1046 | Test cond. B (48 hrs); salt solution shall be 5% concentration | | --- | --- | --- | --- |
| End points: (Same as subgroup 2) | | | | | | | |
| Subgroup 6 | | | $\lambda$ = 10 | | | | |
| High temperature life (nonoperating) | 1031 | $T_{stg}$ = 100 °C | | --- | --- | --- | --- |
| End points: | | | | | | | |
| Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -80 Vdc | | $I_{CBO}$ | --- | -22 | mAdc |
| Emitter to base cutoff current | 3061 | Bias cond. D; $V_{EB}$ = -60 Vdc | | $I_{EBO}$ | --- | -22 | mAdc |

See footnote at end of table.



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.

Table II. Group B inspection. - Continued

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| End points: (Cont) | | | | | | | |
| Forward-current transfer ratio | 3076 | $V_{CE}$ = -2 Vdc; $I_C$ = -1.2 Adc, pulsed (see 4.4.4) | | $h_{FE}$ | 30 | 100 | --- |
| Subgroup 7 | | | $\lambda$ = 10 | | | | |
| Steady state operation life | 1026 | $T_{MB}$ = +75 °C min; $P_C$ = 20 W min. | | --- | --- | --- | --- |
| End points: (Same as subgroup 6) | | | | | | | |

1/ Electrical rejects from the same lot may be used for this test.

Table III. Group C inspection.

| Examination or test | MIL-STD-750 | | LTPD | Symbol | Limits | | |
|---|---|---|---|---|---|---|---|
| | Method | Details | | | Min | Max | Unit |
| Subgroup 1 | | | 20 | | | | |
| Barometric pressure, reduced (altitude operation) | 1001 | Normal mounting; Pressure = 8 mm Hg. for 60 sec min; (see 3.4.1) | | --- | --- | --- | --- |
| Measurement during test: Collector to base cutoff current | 3036 | Bias cond. D; $V_{CB}$ = -80 Vdc | | $I_{CBO}$ | --- | 15 | mAdc |
| Thermal resistance (junction to case) | 3151 | | | $\theta_{J-C}$ | --- | 0.5 | °C/W |

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery and the quality assurance provisions for preparation for delivery shall conform to MIL-S-19500.

6. NOTES

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

Custodians:
Army - EL
Navy - SH
Air Force - 11

Review activities:
Army - EL, MU, SM, MI
Navy - SH
Air Force - 11, 17, 85

User activities:
Army - None
Navy - CG, MC, WP
Air Force - None

Preparing activity:
Navy - SH

(Project 5960-2023)



Source: https://assist.dla.mil -- Downloaded: 2026-10-18T12:58Z
Check the source to verify that this is the current version before use.